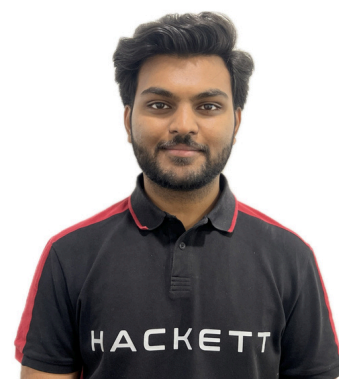


DATE
06/01/2025



VISITOR MANAGEMENT SYSTEM

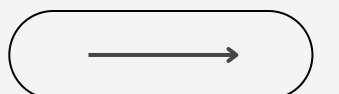


Navadeep Vedantham
navadeep_vedantham@apple.com



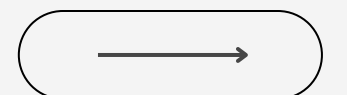
Project Overview

- **Objective:** Streamline visitor check-ins, check-outs, and notifications in a corporate environment.
- **Key Requirements:**
 - Efficient visitor management.
 - Automated notifications for host employees, IT support, and security teams.
 - Scalable and reliable architecture for large office setups.

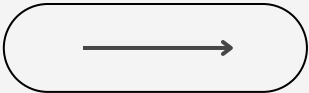
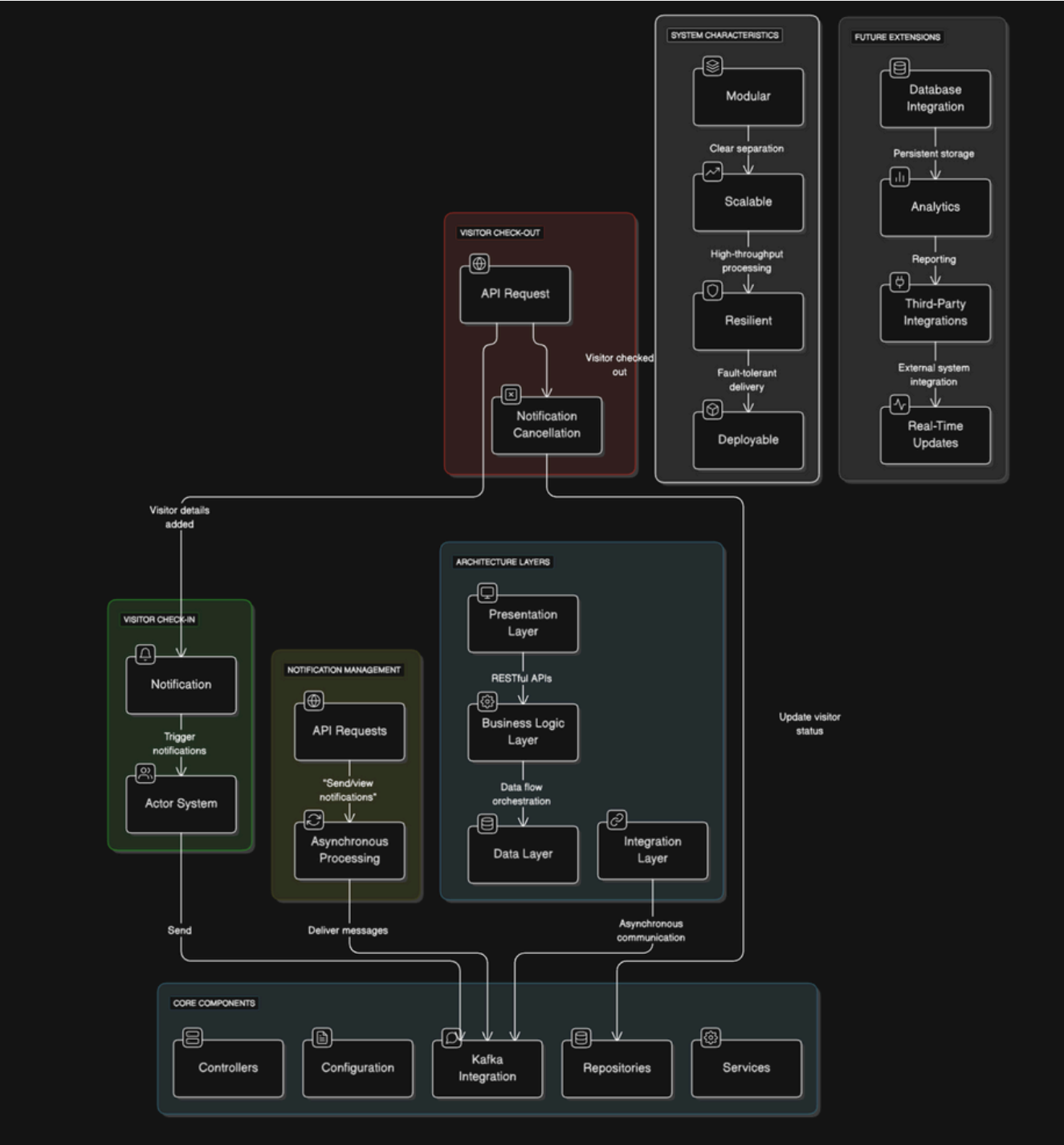
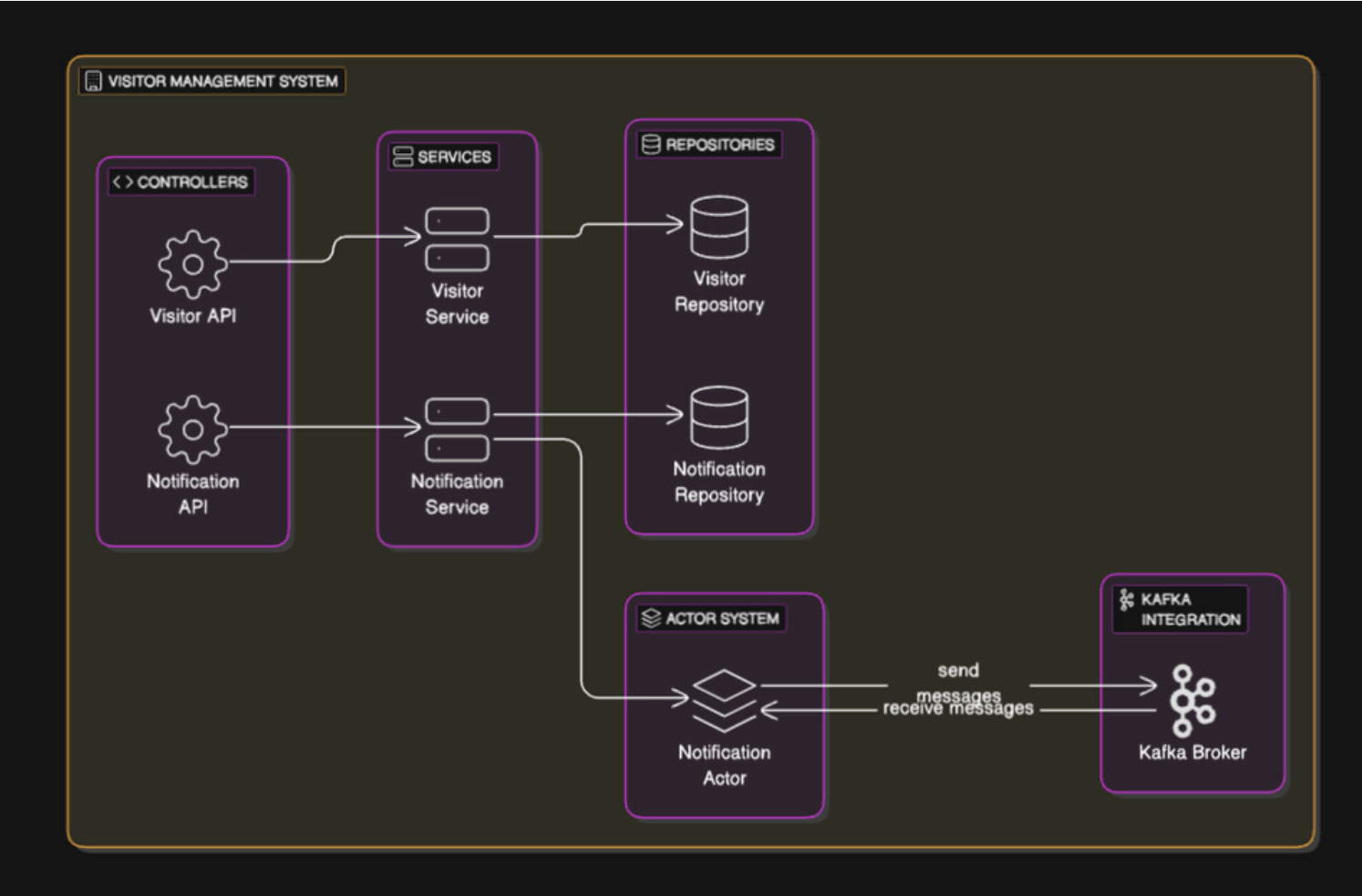


KEY FEATURES

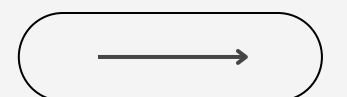
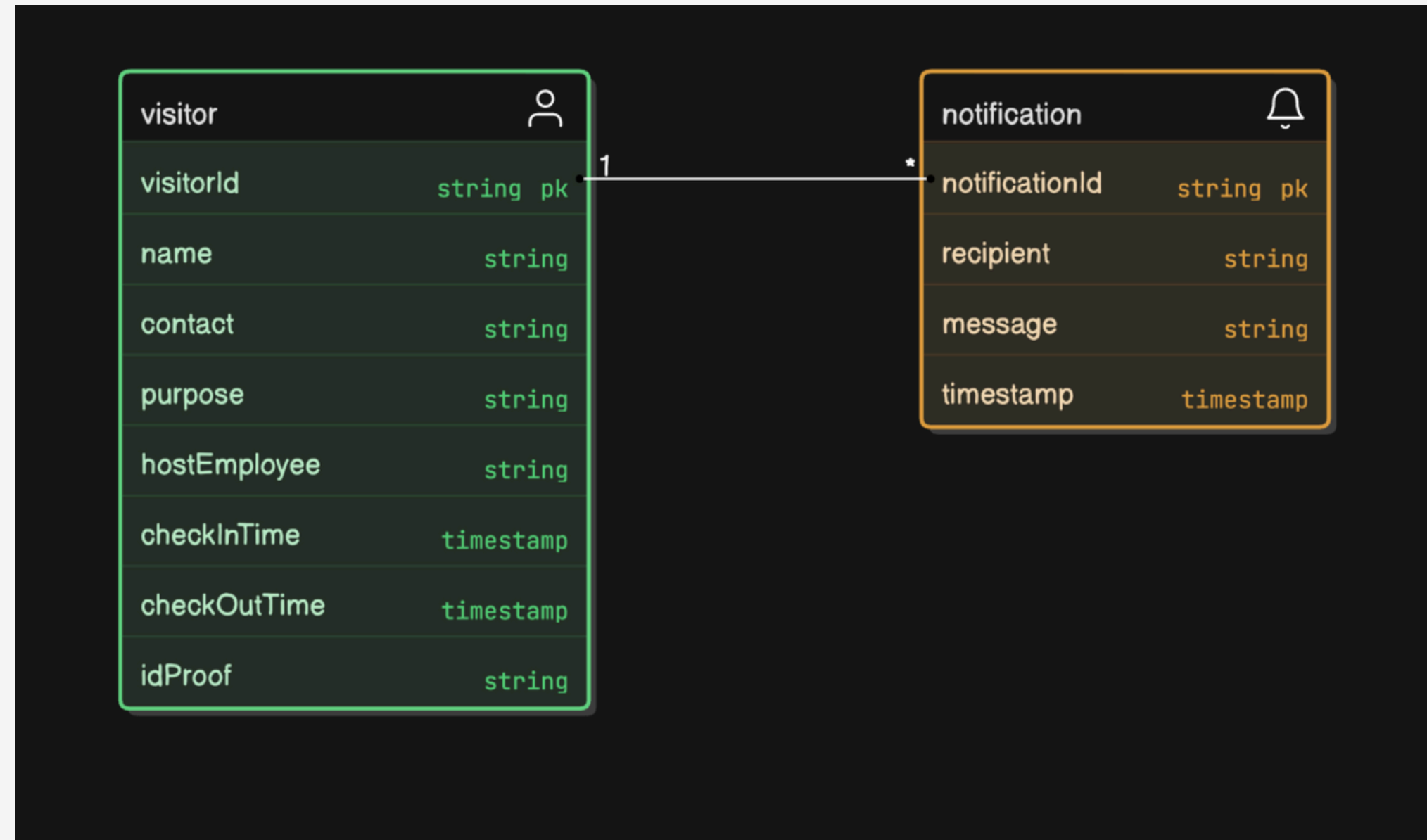
- **Visitor Management:**
 - Add and update visitor details.
 - Track check-ins and check-outs.
 - Delete outdated visitor records.
- **Automated Notifications:**
 - Notify host employees of arrivals.
 - Alert IT for Wi-Fi credential preparation.
 - Inform security for clearance.
 - Cease notifications upon check-out.
- **API-Driven System:**
 - RESTful endpoints for all functionalities.
 - Dockerized Deployment:
 - Consistent environment across all platforms.



SYSTEM ARCHITECTURE

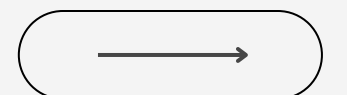


DATA SCHEMA



API Endpoints

- Visitor Management:
 - Retrieve all visitors: GET /visitors.
 - Add a visitor: POST /visitors.
 - Update visitor details: PUT /visitors/:visitorId.
 - Mark visitor check-out: PUT /visitors/:visitorId/checkout.
 - Delete a visitor: DELETE /visitors/:visitorId.
- Notifications:
 - Send new notification: POST /notifications.
 - Retrieve all notifications: GET /notifications.
 - Delete a notification: DELETE /notifications/:notificationId.



MongoDB JSON

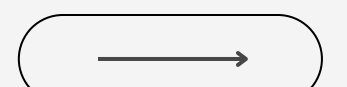
```
{
  "visitors": [
    {
      "_id": "1",
      "name": "John Doe",
      "contactNumber": "1234567890",
      "purpose": "Meeting",
      "checkInTime": "2025-01-06T10:00:00Z",
      "checkOutTime": null
    },
    {
      "_id": "2",
      "name": "Jane Smith",
      "contactNumber": "0987654321",
      "purpose": "Interview",
      "checkInTime": "2025-01-06T09:30:00Z",
      "checkOutTime": "2025-01-06T10:30:00Z"
    }
  ],
  "notifications": [
    {
      "_id": "1",
      "recipient": "John Doe",
      "message": "Your meeting is scheduled at 10:00 AM.",
      "timestamp": "2025-01-06T09:00:00Z",
      "status": "Sent"
    },
    {
      "_id": "2",
      "recipient": "Jane Smith",
      "message": "Your interview is scheduled at 9:30 AM.",
      "timestamp": "2025-01-06T08:30:00Z",
      "status": "Sent"
    }
  ]
}
```

API Responses

```
{
  "getAllVisitors": {
    "method": "GET",
    "url": "/visitors",
    "response": [
      {
        "_id": "1",
        "name": "John Doe",
        "contactNumber": "1234567890",
        "purpose": "Meeting",
        "checkInTime": "2025-01-06T10:00:00Z",
        "checkOutTime": null
      },
      {
        "_id": "2",
        "name": "Jane Smith",
        "contactNumber": "0987654321",
        "purpose": "Interview",
        "checkInTime": "2025-01-06T09:30:00Z",
        "checkOutTime": "2025-01-06T10:30:00Z"
      }
    ]
  },
  "getVisitorById": {
    "method": "GET",
    "url": "/visitors/{visitorId}",
    "example_url": "/visitors/1",
    "response": {
      "_id": "1",
      "name": "John Doe",
      "contactNumber": "1234567890",
      "purpose": "Meeting",
      "checkInTime": "2025-01-06T10:00:00Z",
      "checkOutTime": null
    }
  },
  "addVisitor": {
    "method": "POST",
    "url": "/visitors",
    "request": {
      "_id": "3",
      "name": "Alice Johnson",
      "contactNumber": "9876543210",
      "purpose": "Consultation",
      "checkInTime": "2025-01-06T11:00:00Z"
    },
    "response": {
      "message": "Visitor Alice Johnson added successfully."
    }
  },
  "updateVisitor": {
    "method": "PUT",
    "url": "/visitors/{visitorId}",
    "example_url": "/visitors/1",
    "request": {
      "name": "John Doe Updated",
      "contactNumber": "1234567890",
      "purpose": "Updated Meeting",
      "checkOutTime": "2025-01-06T11:00:00Z"
    },
    "response": {
      "message": "Visitor with ID 1 updated successfully."
    }
  },
  "deleteVisitor": {
    "method": "DELETE",
    "url": "/visitors/{visitorId}",
    "example_url": "/visitors/1",
    "response": {
      "message": "Visitor with ID 1 deleted successfully."
    }
  },
  "getAllNotifications": {
    "method": "GET",
    "url": "/notifications",
    "response": [
      {
        "_id": "1",
        "recipient": "John Doe",
        "message": "Your meeting is scheduled at 10:00 AM.",
        "timestamp": "2025-01-06T09:00:00Z",
        "status": "Sent"
      },
      {
        "_id": "2",
        "recipient": "Jane Smith",
        "message": "Your interview is scheduled at 9:30 AM.",
        "timestamp": "2025-01-06T08:30:00Z",
        "status": "Sent"
      }
    ]
  },
  "getNotificationById": {
    "method": "GET",
    "url": "/notifications/{notificationId}",
    "example_url": "/notifications/1",
    "response": {
      "_id": "1",
      "recipient": "John Doe",
      "message": "Your meeting is scheduled at 10:00 AM.",
      "timestamp": "2025-01-06T09:00:00Z",
      "status": "Sent"
    }
  },
  "sendNotification": {
    "method": "POST",
    "url": "/notifications",
    "request": {
      "_id": "3",
      "recipient": "Alice Johnson",
      "message": "Your consultation is scheduled at 11:00 AM.",
      "timestamp": "2025-01-06T10:30:00Z",
      "status": "Sent"
    },
    "response": {
      "message": "Notification sent successfully to Alice Johnson."
    }
  },
  "deleteNotification": {
    "method": "DELETE",
    "url": "/notifications/{notificationId}",
    "example_url": "/notifications/1",
    "response": {
      "message": "Notification with ID 1 deleted successfully."
    }
  }
}
```


FUTURE ENHANCEMENTS

- Integrate facial recognition for check-ins.
- Add visitor self-service kiosks.
- Implement analytics for visitor flow and trends.



Thank You